



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to T.K.S. FUELS STATION

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

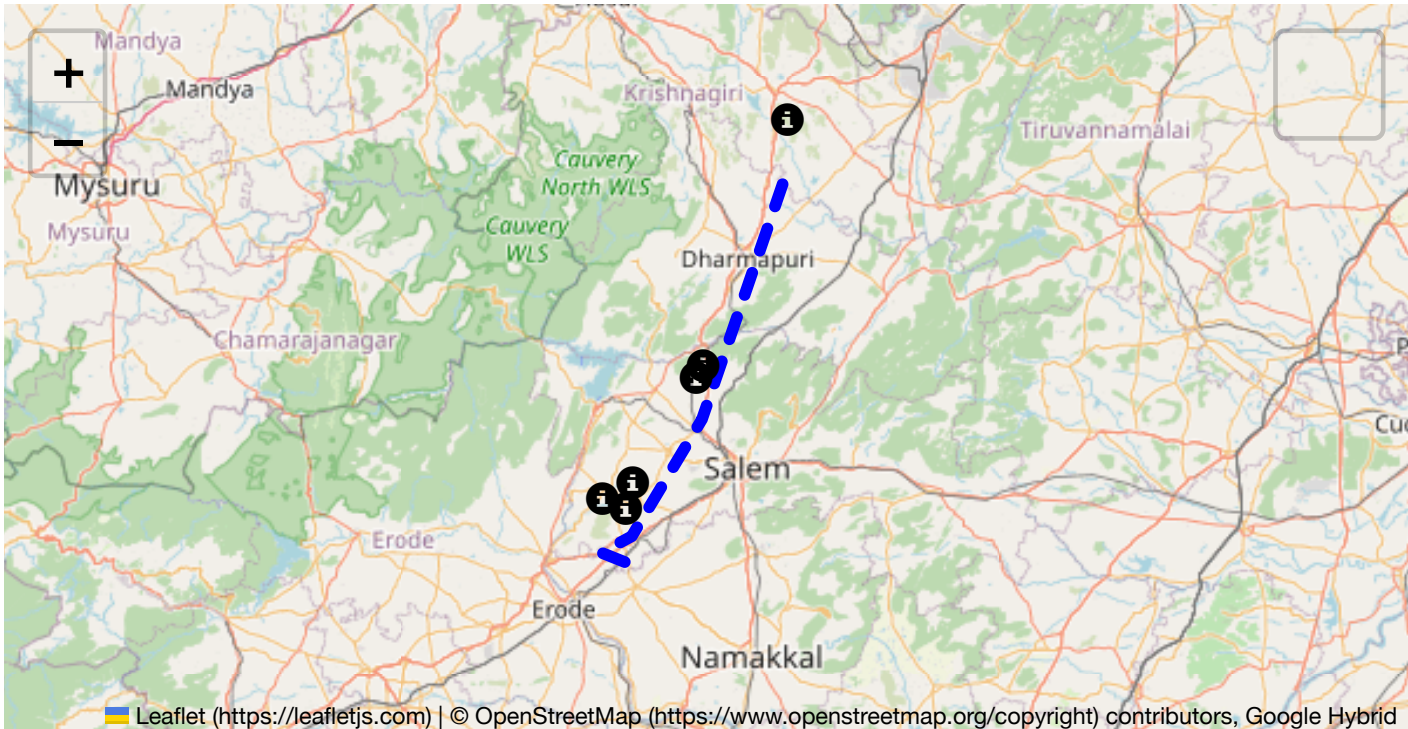
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

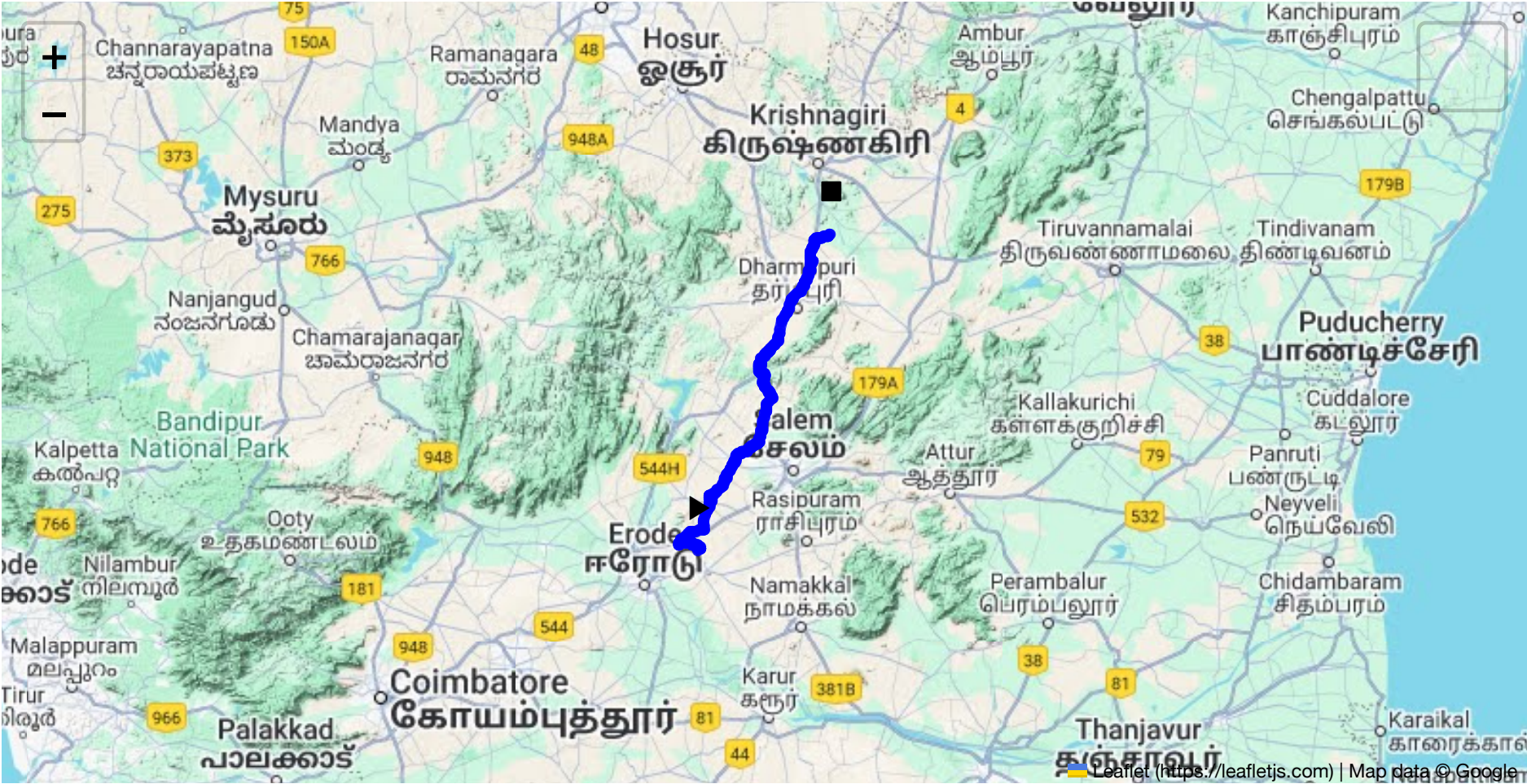
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 132.52 km
Estimated Duration: 2.6 hours
Adjusted Duration (Heavy Vehicle): 3.3 hours

Start: (11.4381, 77.8734)
End: (12.328031, 78.250788)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to S No 356/1C3B, Pannandur Rd traverses through varying terrains and several localities in Tamil Nadu. Starting from Sangagiri, the route leads north towards Bhavani Main Road, subsequently connecting through Ayveli in SH 86 and passing through Salem. It further stretches towards Omalur before finally reaching the destination at Thattarahalli. This route primarily covers state highways and involves navigating urban, semi-urban, and rural settings.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate with monsoons influencing weather patterns. Between October to December, the region often experiences heavy rainfall due to the northeast monsoon. In such conditions, flooding and reduced visibility might pose significant driving hazards. Summers can be intensely hot, with temperatures rising above 40°C, potentially leading to road surface damage and overheated engines.

3. Analysis of Traffic Patterns

Salem, being a major city en route, experiences significant traffic congestion, particularly during peak hours (8-10 AM and 5-8 PM). The roads around Omalur and other small towns may also face congestion during market hours or local events. Rural sections between these urban areas usually have lesser traffic but occasional slow-moving vehicles like tractors or animal-driven carts can cause delays.

4. Assessment of Road Quality and Infrastructure

The state highways (SH 86) are generally in good condition but may have sudden patches of deterioration due to seasonal wear. Urban roads in Salem are typically well-maintained, but construction or repair work

might occasionally obstruct traffic. In rural areas, be aware of poorly-lit sections, and be cautious of stray livestock.

5. Suggestions for Alternative Routes for Emergencies

In the event of a major blockade or emergency, a viable alternative would be to use the NH44, which runs parallel to some sections of the intended route and offers larger lanes suitable for heavy vehicles.

Diversions onto NH544 can also be considered if there are disruptions on the main route near Salem.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu enforces strict regulations on the transportation of hazardous materials, requiring permits and adherence to safety protocols such as prominently displaying necessary signage. Transportation during night hours might be restricted in populated areas to minimize risks.

7. Overview of Historical Incidents

There have been few notable incidents involving heavy vehicles on Tamil Nadu's highways, predominantly related to driver error or mechanical failure. No significant accidents involving hazardous materials have been recently reported, but caution is advised in high-traffic zones like Salem.

8. Environmental Considerations and Sensitive Areas

The route passes through areas with agricultural activities; thus, accidents involving hazardous materials could adversely affect local farmland. Areas near Salem and some rural sections might also be proximate to environmentally sensitive zones, necessitating cautious driving to avoid ecological disturbance.

9. Analysis of Communication Coverage

Cellular coverage is generally reliable along the entire route, although rural stretches might experience brief connectivity issues. It is advisable to carry alternate communication means, especially during emergencies.

10. Estimated Emergency Response Times

Response times can vary; in urban parts such as Salem, emergency services can be expected within 15-30 minutes. In more rural areas, it may extend to 45 minutes or more due to less accessible roads.

12. Overall Summary of Risk Assessment

In summary, this route presents moderate risks mainly associated with traffic congestion, weather conditions, and rural road quality. Truck drivers should be prepared for potential delays and ensure their vehicles are in top condition to handle varied terrains and climates. Adherence to regulations and clear communication with local authorities is essential when transporting hazardous materials. Environmental sensitivity and community impact should be considered as part of the risk management strategy. Regular updates on road and weather conditions will further mitigate unforeseen risks.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45353, 77.85697	15 KM/Hr
4	Turn	Medium	11.45781, 77.85203	30 KM/Hr
5	Turn	High	11.46307, 77.84941	15 KM/Hr
6	Turn	High	11.45542, 77.81725	15 KM/Hr
7	Turn	High	11.45605, 77.81683	15 KM/Hr
8	Turn	Medium	11.49383, 77.88501	30 KM/Hr
9	Blind Spot	Blind Spot	11.49400, 77.88538	10 KM/Hr
10	Turn	High	11.49469, 77.88488	15 KM/Hr
11	Turn	High	11.55618, 77.89789	15 KM/Hr
12	Turn	Medium	11.55629, 77.89888	30 KM/Hr
13	Turn	High	11.58228, 77.90841	15 KM/Hr
14	Turn	High	11.58340, 77.90637	15 KM/Hr
15	Turn	Medium	11.59201, 77.90649	30 KM/Hr
16	Turn	Medium	11.63882, 77.94859	30 KM/Hr
17	Turn	Medium	11.64161, 77.95177	30 KM/Hr
18	Turn	Medium	11.64226, 77.95200	30 KM/Hr
19	Turn	Medium	11.64716, 77.95118	30 KM/Hr
20	Turn	Medium	11.67868, 77.96836	30 KM/Hr
21	Turn	Medium	11.70077, 77.97957	30 KM/Hr
22	Turn	High	11.70223, 77.97860	15 KM/Hr
23	Turn	Medium	11.71402, 77.99586	30 KM/Hr
24	Turn	Medium	11.71704, 77.99948	30 KM/Hr
25	Turn	Medium	11.71918, 78.01127	30 KM/Hr
26	Turn	Medium	11.73769, 78.03451	30 KM/Hr
27	Turn	Medium	11.73785, 78.03551	30 KM/Hr
28	Turn	High	11.73651, 78.03871	15 KM/Hr
29	Turn	Medium	11.74271, 78.04717	30 KM/Hr
30	Turn	Medium	11.86763, 78.08336	30 KM/Hr
31	Turn	Medium	11.87000, 78.08385	30 KM/Hr
32	Turn	Medium	11.88060, 78.08128	30 KM/Hr
33	Turn	Medium	11.91329, 78.05861	30 KM/Hr
34	Turn	Medium	11.92540, 78.06494	30 KM/Hr
35	Turn	Medium	11.95482, 78.04687	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
36	Turn	Medium	11.95633, 78.04755	30 KM/Hr
37	Turn	High	11.95837, 78.04986	15 KM/Hr
38	Turn	Medium	11.96054, 78.04953	30 KM/Hr
39	Turn	High	11.96347, 78.05142	15 KM/Hr
40	Turn	Medium	11.96415, 78.05126	30 KM/Hr
41	Turn	Medium	11.96519, 78.04958	30 KM/Hr
42	Turn	High	11.96620, 78.04920	15 KM/Hr
43	Turn	Medium	11.96842, 78.05097	30 KM/Hr
44	Turn	Medium	12.24648, 78.19537	30 KM/Hr
45	Turn	Medium	12.25238, 78.20039	30 KM/Hr
46	Turn	High	12.31467, 78.20768	15 KM/Hr
47	Turn	Medium	12.31791, 78.21461	30 KM/Hr
48	Turn	High	12.31908, 78.21586	15 KM/Hr
49	Turn	High	12.32377, 78.24215	15 KM/Hr
50	Turn	Medium	12.32526, 78.24360	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
2	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
3	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
4	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
5	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
6	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
7	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
8	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
10	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
11	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
12	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
13	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
14	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
15	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
16	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
17	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
18	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
19	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
20	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
21	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
23	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
25	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
26	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
27	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
29	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
9	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
22	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
24	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
28	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium
30	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45781, 77.85203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45605, 77.81683



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49383, 77.88501



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49400, 77.88538



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49469, 77.88488



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.55618, 77.89789



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58340, 77.90637



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59201, 77.90649



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63882, 77.94859



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64161, 77.95177



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64226, 77.95200



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64716, 77.95118



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67868, 77.96836



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.70077, 77.97957



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70223, 77.97860



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71402, 77.99586



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71704, 77.99948



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71918, 78.01127



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73769, 78.03451



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73785, 78.03551



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73651, 78.03871



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74271, 78.04717



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86763, 78.08336



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87000, 78.08385



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.88060, 78.08128



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91329, 78.05861



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92540, 78.06494



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95482, 78.04687



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95633, 78.04755



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95837, 78.04986



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96054, 78.04953



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96347, 78.05142



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96415, 78.05126



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96519, 78.04958



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96620, 78.04920



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96842, 78.05097



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.24648, 78.19537



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25238, 78.20039



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.31467, 78.20768



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.31791, 78.21461



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.31908, 78.21586



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.32377, 78.24215

